

WASH N PRESS

A Women-Led, Water-Conscious Community Laundry Subscription Platform

Detailed Project Report (DPR) & Functional Requirements Document — Version 1.0

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PART A — DETAILED PROJECT REPORT

1. Executive Summary

Wash N Press is a hyperlocal, subscription-based laundry platform serving gated communities, apartment societies, and PG/co-living accommodations. The model combines app-based convenience for residents with a decentralized, community-embedded operating structure: each serviced society's micro-laundry unit is owned and run by a two-member women's team recruited from the local community, supported by Wash N Press through equipment, training, and working-capital assistance.

The project is built around three pillars: convenience (predictable subscription-based laundry service with live tracking), livelihood creation (women's micro-entrepreneurship at the community level), and sustainability (low-water and water-recycling washing machinery to reduce the environmental footprint of a traditionally water-intensive service).

2. Project Background & Rationale

Urban housing societies in India represent a dense, underserved market for recurring household services. Laundry remains a daily time burden for residents, while existing solutions are either informal (unreliable, no quality assurance) or centralized commercial laundromats (logistics-heavy, impersonal). Wash N Press addresses this gap by embedding the service physically inside the community it serves.

Simultaneously, urban and peri-urban women — particularly those with limited access to formal employment due to caregiving responsibilities, mobility constraints, or lack of formal qualifications — represent an underutilized workforce that benefits from hyperlocal, flexible micro-enterprise opportunities within walking distance of home. This project is structured to address both gaps with one operating model.

3. Project Objectives

- Deliver a reliable, app-based laundry subscription service to residents of partnered housing societies.
- Establish a women-led micro-enterprise (two-member unit) in each serviced society, providing equipment, training, and initial working-capital support.
- Reduce fresh-water consumption per garment processed through water-saving and water-recycling machinery.
- Build a scalable, replicable operating and franchise-style model that can expand society-by-society and city-by-city without proportional increases in central overhead.
- Create a measurable social and environmental impact track record suitable for impact-investment, CSR partnership, and government MSME/SHG scheme alignment.

4. Social Impact Model — Women-Led Micro-Units

4.1 Structure

Each society-level micro-laundry unit is operated by a two-member women's team, ideally residents of that society or its immediate vicinity, minimizing commute and enabling flexible hours around family responsibilities. Wash N Press functions as the enabling platform: providing the machinery (capex), the app/ops system, training, and an initial operating-capital cushion during ramp-up; the women's team provides labor, day-to-day unit management, and the community trust relationship that's difficult for a centralized vendor to replicate.

4.2 Recruitment, Training & Support

1. Identify and recruit candidates through local women's Self-Help Groups (SHGs), RWA networks, or direct community outreach — prioritizing residents of the target society.
2. Provide a structured training program covering machine operation, fabric handling, the Ops App workflow, basic customer service, and hygiene/safety standards.
3. Provide initial working-capital support (e.g., a guaranteed minimum monthly draw for the first 3–6 months) to de-risk the ramp-up period before subscriber volume stabilizes.
4. Pair each new unit with an experienced mentor unit (once the network has matured) for peer support.

4.3 Revenue-Sharing Model

A blended structure is recommended over a flat wage, to align incentives with service quality and retention: a fixed monthly base draw per unit (covering baseline livelihood needs) plus a percentage share of subscription and add-on revenue above an agreed volume threshold. This mirrors structures used in SHG-linked micro-enterprise and managed-franchise models, and should be finalized in consultation with a legal/financial advisor alongside applicable schemes such as India's National Rural Livelihood Mission (NRLM/DAY) or state-level women's enterprise support programs, where the unit may qualify for additional subsidy or low-interest credit support.

4.4 Governance & Accountability

- Each unit operates under a written partnership agreement defining equipment ownership, revenue share, quality standards, and renewal/exit terms.
- Unit performance (quality scores, complaint rate, turnaround time) is tracked through the same Ops App and Admin Console used for centralized hubs, ensuring consistent service standards across the network.
- A grievance and support channel is maintained for the women's teams themselves, separate from the resident-facing support system.

5. Sustainability Model — Water Conservation

5.1 Rationale

Laundry is among the more water-intensive household services, and centralizing or decentralizing that water use without addressing efficiency simply relocates the environmental cost rather than reducing it. Building water efficiency into the core machinery choice — rather than treating it as an optional upgrade — makes sustainability a structural feature of the business rather than a marketing claim.

5.2 Machinery Approach

- Use commercial washing machines with built-in greywater recycling/reuse modules, which capture and filter rinse-cycle water for reuse in subsequent wash cycles, reducing fresh-water draw per load.
- Evaluate low-water-consumption machine models (front-load, high-efficiency commercial units) over conventional top-load machines for all new unit installations.
- Where feasible at higher-volume hubs, consider on-site greywater treatment for reuse in non-potable applications (e.g., facility cleaning), subject to local water-authority approval.

5.3 Economic Framing

Water-efficient machinery should be evaluated as a capex decision with a defined payback period — higher upfront equipment cost weighed against monthly water (and associated heating/pumping) savings — rather than imposed uniformly regardless of unit economics. High-volume hubs, where water cost is a larger share of operating expense, are the natural first deployment point; the payback period should improve as machine costs fall with scale, informing the rollout sequence to smaller micro-units.

5.4 Impact Tracking

- Track and report liters of fresh water saved per garment / per unit / network-wide, as a core sustainability KPI alongside financial metrics.
- Use this data point in RWA pitches, CSR partnership conversations, and impact-investor reporting.

6. Business Model (Revised with Social & Sustainability Layer)

6.1 Revenue Streams

- Subscription fees (Basic / Standard / Premium / Family Pack tiers) — primary revenue stream.
- Pay-per-use for non-subscribers and add-on services (dry cleaning, express delivery, shoe/bag cleaning).
- B2B2C society contracts via RWA tie-ups, with group discount and minimum-volume guarantees.

6.2 Cost Structure

Cost Category	Components	Notes
Processing	Water, power, detergent, equipment depreciation, ironing labor	Reduced water cost where recycling machinery deployed
Women's-unit revenue share	Base draw + performance-linked share	Replaces traditional fixed logistics labor cost at micro-unit level

Cost Category	Components	Notes
Equipment capex	Washing/drying/ironing machinery incl. water-recycling modules	Higher upfront cost for water-efficient models; evaluated on payback period
Logistics (hub model only)	Pickup/delivery staff, routing, fuel	Reduced where micro-units serve their own society directly
Garment risk	Loss/damage claims	Mitigated via QR/RFID batch tagging (see Functional Requirements)
Training & onboarding	Women's-unit recruitment, training program delivery	One-time cost per new unit; amortized over unit lifetime

6.3 Capital Structure Options

- Direct capex funding of unit machinery by Wash N Press, recovered through revenue share over the partnership term.
- Blended funding combining Wash N Press capital with SHG/MSME scheme-linked subsidy or low-interest credit available to women-led micro-enterprises.
- CSR or impact-investment co-funding, given the dual social (livelihood) and environmental (water conservation) impact thesis — a stronger fit for grant or concessional capital than a pure-commercial model.

7. Implementation Roadmap

Phase	Focus	Key Milestones
Phase 1: Pilot (Months 1–6)	1–3 societies in a single city	Recruit & train first women's units; install water-efficient machinery; validate unit economics, garment-loss rate, and resident satisfaction
Phase 2: Hyperlocal Density (Months 6–18)	50–100 societies in 2–3 cities	Standardize training/onboarding playbook; formalize revenue-share and partnership agreement template; build CSR/impact-investor case using Phase 1 data
Phase 3: Cluster Expansion (Year 2+)	Cluster-based density to reduce per-unit onboarding cost	Explore scheme-linked financing (NRLM/SHG); evaluate centralized hub model for very high-density clusters where micro-units have hit volume ceiling
Phase 4: Scale & Brand (Year 3+)	Multi-city, franchise-style replication	Mature mentor-unit network; publish annual social & environmental impact report

8. Risk Analysis

Risk	Mitigation
Slow ramp-up volume at new units affecting women's-team income stability	Guaranteed base draw during initial months; phased rollout tied to pre-validated subscriber interest before unit launch
Quality inconsistency across decentralized units	Standardized training, Ops App-enforced QC checkpoints, network-wide performance tracking
Garment loss/damage disputes	Mandatory QR/Rfid batch tagging from pickup through delivery (see Functional Requirements, Section 4.4)
RWA onboarding delays (committee approval cycles)	Parallel direct-to-resident interest capture; free-trial offers to build resident demand pressure on committees
Higher capex for water-efficient machinery straining early unit economics	Phased rollout starting at higher-volume sites where payback period is shortest
Dependency on scheme-linked financing that may not materialize	Treat as upside, not baseline assumption; core financial model should work on Wash N Press capital alone

9. Expected Social & Environmental Outcomes

- Direct livelihood creation for two women per serviced society, scaling linearly with society count.
- Reduced fresh-water consumption per garment processed versus conventional laundry machinery, tracked and reported network-wide.
- Community-embedded service trust, supporting higher resident retention than a centralized, anonymous vendor relationship.
- A replicable model suitable for partnership with CSR programs, state women's enterprise schemes, and impact-focused investors.

PART B — FUNCTIONAL REQUIREMENTS DOCUMENT

10. Document Purpose & Scope

This section defines the features, user roles, workflows, and system behavior required to build the Wash N Press platform, updated to reflect the women-led micro-unit operating model and sustainability tracking requirements introduced in Part A.

10.1 In Scope

- Resident-facing mobile app (subscriptions, scheduling, tracking, payments)
- Operations app for women's-unit operators, pickup/delivery staff (hub model), and hub workers
- Admin console for society management, women's-unit management, vendor operations, sustainability tracking, and business reporting
- Garment-level tracking system (QR/RFID) to mitigate loss/damage risk
- Society (RWA) onboarding and B2B2C contract workflows
- Women's-unit recruitment, training, and revenue-share management workflows

10.2 Out of Scope (Phase 1)

- Smart locker / IoT hardware integration (Phase 2+)
- Franchise management module (Phase 3)
- Home cleaning bundle and wardrobe management features (future expansion)

11. User Roles & Personas

Role	Description	Primary Interface
Resident / Subscriber	End customer living in a partnered society; subscribes to plans, schedules pickups	Resident App
Non-Subscriber (Guest)	One-off / pay-per-use customer in a serviced area	Resident App (guest checkout)
Women's-Unit Operator (2 per unit)	Operates the in-society micro-laundry unit: processing, light pickup/delivery, resident interface	Ops App (unit mode)
Pickup & Delivery Agent (hub model)	Field staff collecting/returning garments on hub-served routes	Ops App (route mode)
Hub/Facility Worker	Processes garments at a centralized hub (high-density clusters)	Ops App (hub mode)
Society Admin (RWA)	Manages community-level contract, views usage, communicates with residents	Admin Console (society view)
Wash N Press Ops/Impact Manager	Manages units, training pipeline, revenue-share, sustainability KPIs	Admin Console (full access)

Role	Description	Primary Interface
Customer Support Agent	Handles complaints, refunds, garment disputes	Admin Console (support view)

12. Resident App — Functional Requirements

12.1 Onboarding & Account Setup

5. Select society from a list (geofenced/searchable); sign up via mobile number + OTP.
6. Enter flat/unit number, validated against society's registered unit list.
7. Complete profile: name, alternate contact, preferred pickup time windows.

12.2 Subscription Management

- Browse/compare tiers (Basic/Standard/Premium/Family Pack) with garment caps, turnaround, price.
- Select plan, billing cycle, pay via integrated gateway; view usage vs. cap.
- Upgrade/downgrade (next-cycle effective, admin-configurable proration), pause, cancel with reason capture.

12.3 Pickup Scheduling

- View available slots for their society (generated based on the local women's-unit or hub route capacity).
- Book one-time or recurring pickups; reschedule/cancel up to a configurable cutoff.
- Push + SMS/WhatsApp confirmation for every scheduled pickup.

12.4 Order & Garment Tracking

- Unique Order ID per pickup; garment count/category logged by operator at pickup, QR-tagged.
- Live status timeline: Picked Up → At Unit/Hub → In Wash → Ironing/QC → Out for Delivery → Delivered.
- Digital itemized garment receipt for every order, basis for any disputes.

12.5 Wallet, Billing & Payments

- In-app wallet, auto-renewal via saved payment method, full invoice history, UPI/card/net banking support.

12.6 Add-on Services

- Dry cleaning, shoe/bag cleaning, express delivery; special-instructions field per order.

12.7 Support & Issue Resolution

- Raise complaint per order with photo upload; SLA-tracked resolution; auto wallet credit per configured rules.

12.8 Sustainability Visibility (New)

- Optional resident-facing impact widget: "X liters of water saved this month through our recycling machinery," reinforcing the platform's sustainability positioning.

13. Operations App — Functional Requirements

13.1 Women's-Unit Operator Workflow (Primary Model)

8. Operator logs in and views the day's bookings for their society (no multi-society routing needed — single-society scope).
9. At pickup (in-society, short-distance), operator logs garment count/category and applies QR batch tag.
10. Garments processed on-site: Wash → Dry → Iron → QC, each stage logged in-app.
11. QC fail/damage triggers automatic resident notification and support ticket — never silently proceeds.
12. Operator scans QR at delivery to confirm match against pickup manifest; captures delivery confirmation (OTP/signature).
13. App logs water usage per cycle (where machinery supports automated reporting) or allows manual log entry, feeding the sustainability dashboard.

13.2 Hub & Route Model (High-Density Clusters Only)

- Used only where a society/cluster has outgrown micro-unit capacity; retains the route-batching and agent-delivery workflow from the original hub model.
- Same QR-tagging, QC-checkpoint, and discrepancy-blocking logic applies regardless of operating model.

13.3 Inventory & Garment Tracking

- QR (Phase 1) or RFID (Phase 2, pending pilot loss-rate data) batch tag is the system of record from pickup through delivery.
- Lost/damaged garment reporting workflow with photo evidence, linked to order and resident for compensation.

13.4 Women's-Unit Operations Support

- In-app access to training materials and refresher modules.
- Earnings dashboard showing base draw + performance-linked share, updated per billing cycle.
- Direct escalation channel to the Wash N Press Ops/Impact Manager for equipment issues or operational support.

14. Admin Console — Functional Requirements

14.1 Society (RWA) Onboarding

14. Create society profile: name, address, unit structure, contract terms (discount %, minimum volume, exclusivity period).
15. Bulk/manual upload of registered units for resident verification.
16. Configure pickup slots and capacity based on the assigned women's-unit or hub route.

14.2 Women's-Unit Management (New)

- Unit profile: location, operators (2 per unit), equipment installed (incl. water-recycling specifications), training completion status.
- Revenue-share configuration per unit: base draw amount, performance threshold, share percentage above threshold.
- Recruitment pipeline tracker: candidate sourcing (SHG/RWA/direct), interview, training, go-live status.
- Unit performance dashboard: quality score, complaint rate, turnaround time, resident satisfaction — comparable across units for network-wide quality consistency.

14.3 Subscription & Pricing Configuration

- Define tiers, garment caps, pricing — globally or per-society for negotiated B2B rates.
- Configure add-ons, proration/pause/cancellation rules, promotional codes.

14.4 Operations Dashboard

- Real-time order status across all units/hubs; unit and hub capacity/utilization monitoring.
- Route performance (hub model) and garment loss/damage rate tracking by unit/hub/operator.

14.5 Sustainability Dashboard (New)

- Network-wide and per-unit water usage tracking: liters consumed vs. liters saved through recycling machinery, benchmarked against conventional-machine baseline.
- Exportable sustainability impact report for CSR partner and impact-investor reporting.

14.6 Business Reporting & Analytics

Report	Key Metrics	Frequency
Subscription performance	New subs, churn, upgrade/downgrade trends	Weekly/Monthly
Revenue	MRR, ARPU, add-on revenue share	Monthly
Unit/hub economics	Cost per garment, breakeven household count, utilization %, revenue-share payout	Monthly
Social impact	Number of active women's units, livelihoods supported, retention rate of operators	Monthly
Sustainability	Liters of water saved network-wide and per unit	Monthly
Garment risk	Loss/damage incidents per 1,000 garments processed	Weekly

Report	Key Metrics	Frequency
Support SLA	Avg. resolution time, complaint volume by category	Weekly

14.7 Customer Support Tools

- Unified ticket queue across channels; order-level detail view (manifest, photos, unit/hub history).
- Refund/credit issuance to resident wallet with configurable approval thresholds.

14.8 User & Role Management

- Role-based access for admin, ops/impact manager, support agent, and society-level RWA viewer accounts.
- Audit log of all configuration changes (pricing, society contracts, revenue-share terms, refund overrides).

15. Non-Functional Requirements

Category	Requirement
Performance	App screens load within 2 seconds on 4G; status updates reflected within 30 seconds of operator action
Availability	99.5% uptime for resident-facing app and payment processing
Scalability	Must support scaling from single-society pilot to multi-city, hundreds-of-units operation without redesign
Security	PCI-DSS compliant payment handling; OTP authentication; encrypted resident PII storage
Data Privacy	Compliance with India's Digital Personal Data Protection Act (DPDP)
Localization	English plus at least one regional language at launch (e.g., Hindi), extensible further — important given community-level operator usage
Offline Resilience	Ops App must allow offline logging of pickups/deliveries/processing with sync on reconnect
Usability	Operator-facing Ops App UI designed for varying levels of smartphone/digital literacy, given the community-recruitment model

16. Third-Party Integrations

- Payment Gateway: Razorpay or equivalent (UPI, cards, net banking, autopay/eNACH).
- SMS/WhatsApp Business API for transactional notifications.
- Maps/Navigation for hub-model route optimization and live tracking.
- Push Notifications: Firebase Cloud Messaging or equivalent.
- QR scanning: native camera-based for MVP; RFID integration reserved as a Phase 2 upgrade path.

- Water-usage data capture: manual entry for MVP; direct machine telemetry integration where supported by chosen water-recycling machinery vendor, for Phase 2.

17. End-to-End Order Workflow (Summary)

17. Resident books or has a recurring pickup slot with their society's women's unit.
18. Operator logs garment count, applies QR batch tag, marks pickup complete → resident notified.
19. On-site processing: Wash → Dry → Iron → QC, with status updates at each stage; water usage logged.
20. QC pass moves to "Ready for Delivery"; QC fail triggers resident notification and support ticket.
21. Operator delivers, scans QR to confirm match, captures delivery confirmation.
22. Order closed; garment usage deducted from subscription cap; resident may rate and dispute within a configurable window.

18. Open Items for Product & Operational Decision

- Final revenue-share formula for women's units (base draw amount, threshold, percentage) — requires finance and legal input, and alignment with any SHG/MSME scheme terms.
- QR vs. RFID tagging decision — pending validated loss/damage rate from the Phase 1 pilot.
- Water-recycling machinery vendor selection and payback-period analysis per unit size.
- Threshold at which a society transitions from a women's-unit model to a centralized hub model (very high-density clusters).
- Legal structuring of the women's-unit partnership agreement (employment vs. micro-franchise vs. SHG-linked enterprise) — requires legal review.
- Identification of applicable government schemes (NRLM/DAY, state women's enterprise programs) for potential co-funding or credit support.